NC STATE UNIVERSITY

MSE 360 Course Syllabus

Kinetic Processes in Materials

Spring 2025

INSTRUCTOR INFORMATION

Position	Name	Phone	Email	Office Location
Instructor	Prof. Veronica Augustyn	919.515.3272	vaugust@ncsu.edu	EB I 3070D
Teaching Assistant	Muhammad Hoseini		shosein@ncsu.edu	EB I 3035

Office Hours

Instructor office hours: after class Thursday from 11:30 to 12:30 pm. Other times available by appointment on Zoom.

TA office hours: Monday from 11 am to 12:30 pm. Other times available by appointment on Zoom.

Preferred Method of Communication

The preferred contact method is via e-mail.

Response Time

The typical response time to emails is within 24 hours on weekdays.

COURSE INFORMATION

Course Website: https://wolfware.ncsu.edu/courses/my-wolfware/

Course Credit Hours: 3

Meeting Time

Tuesday & Thursday, 10:15 - 11:30 am, Fitts-Woolard Hall 02115

Prerequisites/Corequisites

MA 341: Applied Differential Equations
MSE 301: Introduction to Thermodynamics of Materials

General Education Program (GEP) Information

none

COURSE OVERVIEW

Catalog Description

Types, mechanisms, and kinetics of solid-state phase transformations are covered with selected applications to all classes of materials. Mechanisms of diffusion and techniques for diffusion calculations are presented. The role of surface energy and strain in the evolution of structure during transformation is presented. Phenomena at different size scales (atomic, nano, micro) are described relative to the evolution of structure during transformation.

Course Structure

This is an in-person, synchronous course. Homework assignments are due by midnight on Saturday. Each week I will send an email with the weekly course plan which will include the topics we will cover in lecture, the reading assignments, and the homework assignment. Reading assignments are **required** - the course textbook is available for free as an e-book from the NC State Library. All assignments will be submitted via Moodle except for exams, which will be in-person in the course classroom.

LEARNING OUTCOMES

Upon completion of this course, students will be able to:

1. Define and describe various mechanisms of diffusion in materials.
2. Solve diffusion-related mathematical problems in materials science.
3. Explain the kinetics of vapor-solid, liquid-solid, and solid-solid phase transformations in materials.
4. Apply the principles of phase transformations to materials engineering processes.
5. Identify and describe effects of bulk, strain, and surface energy in changes of state and structure.
6. Describe diffusional and diffusion-less transformations.

COURSE MATERIALS

Required Textbook

O'Hayre, Ryan. Materials Kinetics Fundamentals: Principles, Processes, and Applications. Hoboken, Wiley,

2015. (available as an e-book from the NC State Library)

Required Software

MATLAB (available from NC State) or other mathematical computing software.

GRADING

Grade Components

Component	Weight	Details
Homework	30%	Individual problem sets for weekly modules. There are 12 homework sets. Each homework is worth 2.5%.
Midterm Exams	54%	3 midterm exams each worth 18% taken during class. Tentative schedule is in the syllabus. Lowest midterm score may be dropped, so the other 2 exams each become worth 27%.
Final Exam	16%	Final exam given during the official final exam time. The final exam is cumulative.

Grading Scale

This course uses standard NC State letter grading:

Low	Letter	High
97 ≤	A+	≤ 100
93 ≤	A	< 97
90 ≤	A-	< 93
87 ≤	B+	< 90
83 ≤	B	< 87
80 ≤	B-	< 83
77 ≤	C+	< 80
73 ≤	C	< 77
70 ≤	C-	< 73
67 ≤	D+	< 70
63≤	D	< 67
60 ≤	D-	< 63
0 ≤	F	< 60

Requirements for Credit-Only (S/U) Grading

To receive a grade of S, students are required to take all exams and quizzes, complete all assignments, and earn a grade of C- or better. Conversion from letter grading to credit only (S/U) grading is subject to university deadlines. Refer to the Registration and Records calendar for deadlines related to grading. For more details refer to http://policies.ncsu.edu/regulation/reg-02-20-15.

Requirements for Auditors (AU)

Information about and requirements for auditing a course can be found at http://policies.ncsu.edu/regulation/reg-02-20-04.

COURSE SCHEDULE

Week	Date Range	Topic	Readings	Activities
Week 1	1/5 - 1/11	Introduction	Ch. 1.1 - 1.7	Homework 1
Week 2	1/12 - 1/18	Kinetics: Homogeneous chemical reactions	Ch. 3.1 - 3.2	Homework 2
Week 3	1/19 - 1/25	Kinetics: Activation theory & Heterogeneous chemical reactions	Ch. 3.3 - 3.5	Homework 3
Week 4	1/26 - 2/1	Diffusion: Flux & forces, Steady state vs. Transient, Fick's 1st law	Ch. 4.1 - 4.2	Study for Exam 1
Week 5	2/2 - 2/8	**Exam 1** & Diffusion: Fick's 2nd law	Ch. 4.4.2	**2/4: Exam 1** (Ch. 1 & 3) Homework 4
Week 6	2/9 - 2/15	Diffusion: Solution to Fick's 2nd law for semi-infinite diffusion	Ch. 4.4.2	**2/11: Wellness Day** (no class) Homework 5
Week 7	2/16 - 2/22	Diffusion: Other solutions to Fick's 2nd law	Ch. 4.4.2	Homework 6
Week 8	2/23 - 3/1	Diffusion: Moving interface, Coupled diffusion	Ch. 4.4.3 - 4.4.5	Homework 7 & study for Exam 2
Week 9	3/2 - 3/8	Diffusion: Atomistic treatment & **Exam 2**	Ch. 4.5 - 4.6	**3/6: Exam 2** (Ch. 4)
Week 10	3/9 - 3/15	**Spring Break**		
Week 11	3/16 - 3/22	Gas-Solid (G-S) Kinetics: Active Gas Corrosion & Chemical Vapor	Ch. 5.1 - 5.3	Homework 8

		Deposition		
Week 12	3/23 – 3/29	G-S Kinetics: Atomic Layer Deposition & Passive Oxidation	Ch. 5.4 - 5.6	Homework 9
Week 13	3/30 - 4/5	Solid-Liquid (S-L) & Solid-Solid (S-S) Phase Transformations: Driving Forces, Spinodal Decomposition, Surfaces & Interfaces	Ch. 6.1 - 6.4	Homework 10
Week 14	4/6 - 4/12	S-L & S-S Phase Transformations: Nucleation & Growth, Solidification & **Exam 3**	Ch. 6.5 - 6.10	**4/10 Exam 3 (Ch. 5 & 6)**
Week 15	4/13 - 4/19	**Exam 3** & Microstructural Evolution: Capillary Forces & Surface Evolution	Ch. 7.1 - 7.2	Homework 11
Week 16	4/20 – 4/26	Microstructural Evolution: Coarsening, Grain Growth, & Sintering	Ch. 7.3 - 7.6	Homework 12 & study for Final Exam
Cumulative Final Exam: Tuesday, 4/29, 8:30 am - 11 am				

Please Note: Course schedule is subject to change.

COURSE POLICIES

Late Assignments

Late assignments will be accepted with 5 points taken off for every day submitted late. Assignments submitted later than two weeks passed the original due date will NOT be accepted.

Incomplete Grades

Standard NC State Policy for Incomplete Grades:

If an extended deadline is not authorized by the instructor or department, an unfinished incomplete grade will automatically change to an F after either (a) the end of the next regular semester in which the student is enrolled (not including summer sessions), or (b) the end of 12 months if the student is not enrolled, whichever is shorter. Incompletes that change to F will count as an attempted course on transcripts. The burden of fulfilling an incomplete grade is the responsibility of the student. The university policy on incomplete grades is located at http://policies.ncsu.edu/regulation/reg-02-50-3.

Attendance and Participation

Students are expected to attend all classes and to take the exams on scheduled exam dates. No make-up exams will be given. The lowest Midterm Exam score (of 3) will be dropped (see "Grade Components" above). Students anticipating several absences during the term should discuss the situation with the Instructor during the first two weeks of classes. See "Late Assignments" policy above.

For complete attendance and excused absence policies, please see NCSU REG 02.20.03.

For information on the withdrawal process and timeline, please see https://studentservices.ncsu.edu/your-classes/withdrawal/process/.

Use of Artificial Intelligence (AI) Tools

This course recognizes the potential of artificial intelligence (AI) tools, such as chatbots, text generators, paraphrasers, summarizers, or solvers, to enhance your learning and creativity. You are welcome to use AI tools as supplementary resources to assist you with your assignments, if you do so in an ethical and responsible manner. This means that you must:

Use AI tools only for tasks that are appropriate for your level of learning and understanding. Do not use AI tools to replace your own thinking or analysis, or to avoid engaging with the course content.

Cite any AI tools you use properly: provide the name of the AI tool, the date of access, the URL of the interface, and the specific prompt or query you used to generate the output.

For example: Bing. "recent high quality instructional materials for teaching algebra to college students." Accessed August 1, 2023. https://www.bing.com/chat.

Provide evidence of how you used the AI tool and how it contributed to your assignment. Explain what you learned from the AI tool, how you verified its accuracy and reliability, how you integrated its output with your own work, and how you acknowledged its limitations and biases.

Take full responsibility for any mistakes or errors made by the AI tool. Do not rely on the AI tool to produce flawless or correct results. Always check and edit the output before submitting your work. If you discover any inaccuracies or inconsistencies in the output after submission, notify the instructor immediately and correct them as soon as possible.

Using AI tools in an unethical or irresponsible manner, such as copying or paraphrasing the output without citation or evidence, using the output as your own work without verification or integration, or using the output to misrepresent your knowledge or skills, is considered a form of academic dishonesty and will result in a zero grade for the assignment and possible disciplinary action. If you have any questions about what constitutes ethical and responsible use of AI tools, please consult with the instructor before submitting your work.

UNIVERSITY POLICIES

Academic Integrity and Honesty

Students are required to comply with the university policy on academic integrity found in the Code of Student Conduct. Therefore, students are required to uphold the university pledge of honor and exercise honesty in completing any assignment.

Please refer to the Academic Integrity web page for a detailed explanation of the University's policies on academic integrity and some of the common understandings related to those policies.

Violations of academic integrity will be handled in accordance with the Student Discipline Procedures (NCSU REG 11.35.02).

Students are responsible for reviewing the NC State University PRR's which pertains to their course rights and responsibilities:

- Equal Opportunity and Non-Discrimination Policy Statement and additional references
- Code of Student Conduct
- Grades and Grade Point Average
- Credit-Only Courses
- Audits

Students with Disabilities

Reasonable accommodations will be made for students with verifiable disabilities. To take advantage of available accommodations, students must register with the Disability Resource Office at Holmes Hall, Suite 304, Campus Box 7509, 919-515-7653. For more information on NC State's policy on working with students with disabilities, please see the Academic Accommodations for Students with Disabilities Regulation (NCSU REG 02.20.01).

Non-Discrimination Policy

NC State provides equal opportunity and affirmative action efforts, and prohibits all forms of unlawful discrimination, harassment, and retaliation ("Prohibited Conduct") that are based upon a person's race, color, religion, sex (including pregnancy), national origin, age (40 or older), disability, gender identity, genetic information, sexual orientation, or veteran status (individually and collectively, "Protected Status"). Additional information as to each Protected Status is included in NCSU REG 04.25.02 (Discrimination, Harassment and Retaliation Complaint Procedure). NC State's policies and regulations covering discrimination, harassment, and retaliation may be accessed at http://policies.ncsu.edu/policy/pol-04-25-05 or https://oied.ncsu.edu/divweb/. Any person who feels that he or she has been the subject of prohibited discrimination, harassment, or retaliation should contact the Office for Equal Opportunity (OEO) at 919-515-3148

Basic Needs Security

Any student who faces challenges securing their food or housing or has other severe adverse experiences and believes this may affect their performance in the course is encouraged to notify the professor if you are comfortable in doing so. Alternatively, you can contact the Division of Academic and Student Affairs to learn more about the Pack Essentials program https://dasa.ncsu.edu/pack-essentials/.

COURSE EVALUATIONS

ClassEval is the end-of-semester survey for students to evaluate instruction of all university classes. The current survey is administered online and includes 12 closed-ended questions and 3 open-ended questions. Deans, department heads, and instructors may add a limited number of their own questions to these 15 common-core questions.

Each semester students' responses are compiled into a ClassEval report for every instructor and class. Instructors use the evaluations to improve instruction and include them in their promotion and tenure dossiers, while department heads use them in annual reviews. The reports are included in instructors' personnel files and are considered confidential.

Online class evaluations will be available for students to complete during the last two weeks of the semester for full semester courses and the last week of shorter sessions. Students will receive an email directing them to a website to complete class evaluations. These become unavailable at 8am on the first day of finals.

- Contact ClassEval Help Desk: classeval@ncsu.edu
- ClassEval website
- More information about ClassEval

SYLLABUS MODIFICATION STATEMENT

Our syllabus represents a flexible agreement. It outlines the topics we will cover and the order we will cover them in. The pace of the class depends on student mastery and interests. Thus, minor changes in the syllabus can occur if we need to slow down or speed up the pace of instruction.

Changes to the syllabus will be communicated in class and via e-mail.